Model Development Report - Part 2

Training and Tuning | Objectives, validation and uncertainty

1. Training results: validation inside training only

EVS subset: 59,438 respondents / 36 countries; A170 has 59,122 valid responses. The fixed split yields 34,740 training, 5,007 production-validation, 10,079 same-country final-test and 9,296 held-out-country final-test cases. The 316 missing targets are excluded from predictive learning. Counts differ slightly from nominal percentages due to per-stratum rounding.

Held-out countries: AT, FR, GB, LT, MK, RS. Neither final prediction test was used for tuning or model comparison. Five within-country folds share country contexts but not respondents; five group folds remove entire training-country contexts.

Approach	Criterion	Fitted summary	Chosen tuning	Optimizer	CV MAE +/- SE Log loss RPS
baseline	Empirical absolute loss / frequencies	Empirical frequencies	None	Closed form	1.565 +/- 0.009 1.999 0.1251
ordinal	Threshold logistic + L2	81 coefficients + 9 cutpoints	alpha=0.01	L-BFGS-B; analytic gradient	1.255 +/- 0.007 1.802 0.1001
forest	Gini split gain	200 trees	depth 14; leaf 10; mtry 0.7	Greedy CART; seed 42	1.250 +/- 0.005 1.753 0.0987
boosting	Multiclass log loss	780 trees	lr 0.1; leaves 15; leaf 50; L2 1.0	Histogram boosting; early stop	1.225 +/- 0.006 1.715 0.0963

Frozen selection: boosting; rule: lowest mean CV MAE. MAE uses the median of the predicted 10-category distribution. SE is the sample SD of five fold MAEs divided by sqrt(5): a descriptive tuning uncertainty, not a survey-design or independent-fold confidence interval.

2. Criteria and representation

Ordinal fitting minimizes all-threshold logistic surrogate loss plus L2; the baseline uses observed frequencies. Forests greedily improve Gini impurity; histogram boosting minimizes multinomial log loss. Final selection instead targets CV MAE; log loss and RPS check probability quality. No class balancing is applied, because it would change the learned prevalence. All predictive fitting is unweighted; survey-weighted inference answers a separate population question.

Age and age squared use fold-specific median imputation and scaling; a missing-age indicator is added. Nominal and ordinal survey items use explicit categories, including missingness. Country holdouts receive unknown-country indicator blocks during group CV; the public command rejects unsupported countries. The production artifact contains only 30 training-country contexts.

3. Alternative split criterion

Fixed forest settings; changed criterion	CV MAE	Log loss	RPS
Gini	1.2504	1.7530	0.09870
Shannon entropy	1.2535	1.7685	0.09876

Changing only the criterion changes CV MAE by +0.0031. This is a controlled comparison of greedy split objectives; it is not evidence that the forest finds a global likelihood optimum. Probability and point-score changes are both reported.

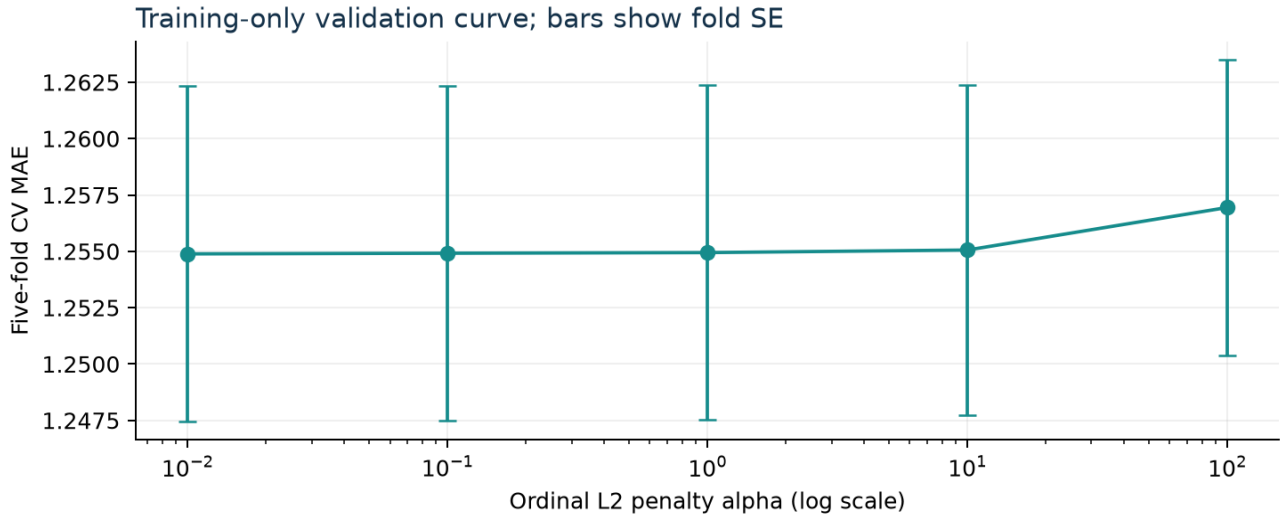
4. Search, chosen hyperparameters and sensitivity

Candidate	Selected hyperparameters
baseline	{}
ordinal	{"alpha": 0.01}
forest	{"max_depth": 14, "max_features": 0.7, "min_samples_leaf": 10}
boosting	{"l2_regularization": 1.0, "learning_rate": 0.1, "max_leaf_nodes": 15, "min_samples_leaf": 50}

Initial search and one-step boundary extensions follow Part I. Every candidate uses five folds and identical split indices; no preprocessing is shared across folds. Forest size is fixed at 200 trees. Boosting uses at most 200 iterations, a 15% internal training-fold validation fraction, patience 15, and seed 42. Complete trial spaces, fold results and timings are in results/trials.json.

Frozen selected-family specification	CV MAE	Log loss	RPS
demographic_economic	1.447	1.892	0.1140
with_happiness	1.102	1.625	0.0870

Sensitivity fits reuse the selected family's hyperparameters without retuning; differences assess predictor availability, not separately optimized competitors. Adding happiness tests near-proxy dependence; removing subjective inputs tests whether survey-like subjective answers are essential.



5. Optimization and fitted parameters

Ordinal: L-BFGS-B with analytic gradients, ordered-threshold increment constraints, maximum 2,000 initial iterations, initial ftol=1e-9; retry at stricter tolerance when necessary. Final fit: 135 iterations; success=True; objective 100772.10; projected gradient per training row 2.8e-05. **Boosting:** final fit 78 iterations / 780 trees; internal validation log loss 1.9991 to 1.7282. Alternate-seed details are recorded in seed_stability.json.

Forests use greedy CART and bootstrap aggregation; there is no continuous gradient convergence test. Boosting uses greedy histogram trees with early stopping, which is part of regularization rather than a certificate of global optimality. A second seed is reported for the chosen tree configurations as an optimizer/randomization stability check. Full settings, thresholds, coefficients, tree counts and diagnostics are saved as JSON; fitted estimators are serialized separately.

6. Transfer diagnostics and uncertainty

Frozen candidate	Grouped CV MAE	Log loss	RPS
baseline	1.565	2.003	0.1254
ordinal	1.282	1.822	0.1022
forest	1.281	1.782	0.1008
boosting	1.266	1.760	0.0997

Group CV is a secondary training-country diagnostic, not nested CV or a score on the six held-out test countries: configurations were chosen using within-country CV across the training sample. It exposes transfer to unfamiliar country indicator blocks. Country/subgroup errors and reliability bins are provided as aggregates. Predicted category probabilities express model uncertainty; no individual prediction intervals are claimed.

7. Population estimates and descriptive gaps

Overall weighting	High-satisfaction proportion	Approx. SE	Approx. 95% interval
unweighted	72.51%	0.18 pp	72.17-72.86%
equal_country	70.99%	0.24 pp	70.55-71.49%
population	68.60%	0.47 pp	67.73-69.58%

Descriptive contrast	Difference	Approx. SE	Approx. 95% interval
income high minus low	+31.74 pp	0.82 pp	+30.06 to +33.31 pp
paid minus unemployed	+18.99 pp	0.97 pp	+17.02 to +20.94 pp

These pre-specified inferential descriptions use all 36 country samples separately from predictive model selection. Within-country gwght adjusts observed margins; pooled equal-country-size and population-size weighting target different summaries. Intervals use 1,000 iid respondent bootstraps within fixed countries. They omit unobserved PSU dependence, do not estimate uncertainty over a hypothetical population of countries, and do not remove selection or measurement bias. Small or low-effective-size groups are suppressed.

Income is missing in 14.70% of EVS rows. Equal-country-size-weighted high-satisfaction proportions are 70.84% with observed income and 71.77% with missing income. This diagnoses a potentially selected complete-case population; it does not identify an MNAR correction.

8. Causality, reproducibility and remaining evaluation

The paid-employment gap above is an association. The effect of obtaining employment remains unidentified: no pre-employment well-being, transition assignment or 12-month outcome is observed. No causal coefficient or causal standard error is reported. The prospective design and assumptions in Part I remain required.

Environment: Python 3.12.14; scikit-learn 1.9.1; NumPy 2.5.4; SciPy 1.18.1. Seed 42. Input Parquet SHA-256: 978592d03407591d74b1... Full source/code hashes, package versions, split policy and optimizer settings are recorded. Source: EVS/WVS ZA7505 v5.0.0, doi:10.4232/1.14320. Reproduce with make train, make inference and make develop-reports.